Planning forest management under uncertain ecological forecasts

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European forests are vulnerable to climate change. Maintaining their integrity will require adapted management practices, but current projections for the future state of these forests are highly variable. Understanding what drives this variability would help decision makers and forest managers. We compared forecasts for nine species of European forest trees across 1,350 combinations of conditions, spanning a range of ecological model types, emission, and climate scenarios. We found that differences between ecological models were the largest source of uncertainty (40 to 64%), surpassing differences between climate models (9 to 32%) and emission scenarios (SSP2 vs. SSP5, 10 to 14 %). Our results show that the inconsistency in ecological model projections is a major current limitation that hinders effective decision-making. However, we identified large areas such as Scandinavia and Southern Europe with relatively consistent model projections where strong managements decisions could be taken. We also identified other areas such as Central Europe where the large divergence between model projections calls for more risk-averse strategies. We demonstrate that considering the different sources of uncertainty can improve decision-making today, until ecological forecasting improves.

Main

Forests are key to climate change mitigation policies and achieving carbon neutrality[1,2]. In Europe, where summer temperatures are rising faster than anywhere on Earth[3], forests are already impacted by unprecedented pulses of tree mortality[4], due to an increase in burned areas[5,6], and a higher frequency of pest and drought-induced diebacks[7–9]. As a result, some forests are becoming net sources of CO_2[10,7].

Forest management can partially mitigate these threats and enhance forest resilience, including by selecting species better adapted to future conditions. Such adaptation requires reliable forecasts of future tree species distributions. Species shifts are expected to have a major impact on timber production and the forest economic sector[11,12]. To preserve the socio-economic functions of forests, managers need to know whether the current species will withstand future climatic conditions, whether they can rely on natural regeneration, or whether they should consider new species opportunities. However, the practical relevance of tree species distribution models for forest management is still limited because these models provide highly divergent projections of species distribution[13–18].

Divergent forecasts are often driven by the underlying approach of the model, whether based on empirical correlations or physiological processes. It remains unclear under which conditions one approach is more reliable than another[19], but most projections still rely on a limited set of models[20,11,12,21]. This masks potentially significant uncertainties and ultimately increases the risk of supporting incorrect policy and management decisions[22]. Addressing this risk requires a better understanding of the nature of uncertainties for decision-making in the forest sector[23–26].

Current projections of tree distributions compound several layers of uncertainty, climatological, socio-economic, and ecological. If most of the variation in the projections of species range

shifts is due to the ecological representation of processes across modelling approaches, even more than global emissions scenarios, it is critical to encompass a wide range of models to assess this uncertainty. Failing to do so could lead to overly confident predictions about which species will or will not survive in future climate, ultimately leading to forest management decisions based on flawed evidence.

To explore this issue, we modeled shifts in forest tree species distributions from 1970 to 2100 across five main European biomes. Projections were performed for nine species of trees, both broadleaf and needleleaf, adapted to various climatic conditions throughout Europe and representing two-thirds of the European forested area (Supplementary Figure 1). Rather than relying on a single modelling approach, our objective was to quantify how differences among ecological and climatological models contribute to uncertainty in projections and how this uncertainty can inform forest management decisions. We incorporated a range of ecological models, mechanistic, correlative and 'hybrid' models. We also considered a range of climate model predictions, using five global climate models with different climate sensitivities, and two emission ('forcing') scenarios, resulting in a set of ten future climatic conditions. Including the different sources of variability in the projections allowed us to quantify the contribution of each source to the total variation between the projections. Overall, we used an ensemble of 1,350 projections to assess their consistency across species and biomes.

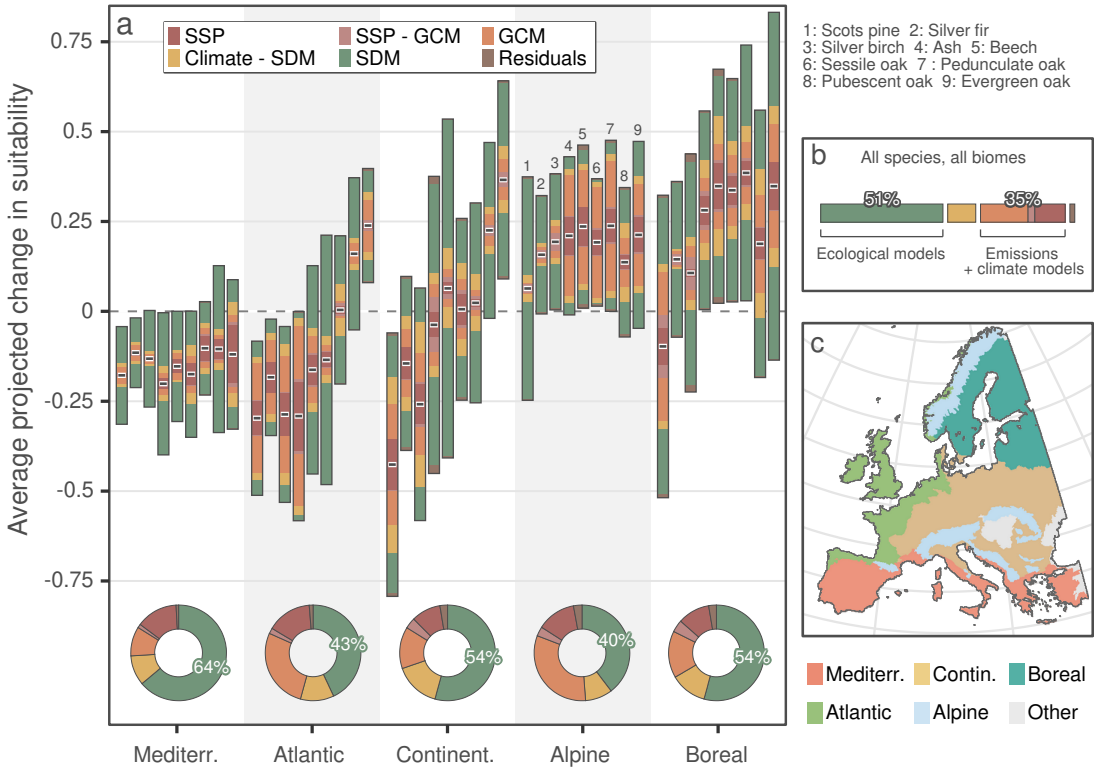


Figure 1: Ecological models represent the main source of uncertainty in future projections of species distribution across European biomes (2080-2100). Level of uncertainty associated with projected changes in climate suitability relative to the historical period (1970–2000), **(a)** for each species within biome and **(b)** across all species and biomes. **(c)** Map of the five main European biomes. Five uncertainty sources are distinguished: (i) the future Shared Socioeconomic Pathways (SSP), (ii) the Global Climate Model (GCM) used for the climate forcing scenario, (iii) the interactions between the SSP and the GCM, (iv) the Species Distribution Model (SDM), and (v) the interactions between SDM and climate projections (both GCMs and SSPs). For each species, the black horizontal line represents the mean projected change in climate suitability. Uncertainty ranges at 90% were calculated additively and symmetrically around the mean.

Results and discussion

Differences between ecological models explained more variation in projected species distributions than differences among climate trajectories. We found that the choice of the ecological model explained the largest proportion of uncertainty (51%) across species and biomes, while the choice of the socioeconomic pathway contributed a smaller portion of the uncertainty (35%, Figure 1b). In the driest biome, the Mediterranean region, a consistent decline in climatic suitability was observed for the nine species in the future, regardless of the ecological model or climatic scenario (Figure 1a). Both the Atlantic and continental biomes showed variable trends between species, demonstrating the importance of making species-specific management plans (Figure 1a). In the Atlantic biome, for example, where sessile oak represents an important cultural and economic value, we found that more than 80% of the uncertainty in the projections was due to the variation between the ecological models. In the Alpine and boreal biomes, an increase in climate suitability was observed for most of the species considered here.

Correlative models underestimate uncertainty

The choice of the type of ecological model had a major influence on the attribution of uncertainty, because much of the forecast variability arises at the level of the ecological models and their parameters (Figure 2). Correlative models alone tend to provide truncated forecasts of historical and future species distributions. Accounting for all types of ecological models led to a wider range of predicted changes in species distribution (Figure 2). This inconsistency between ecological model types reveals the limits in our understanding of species responses to climate change and the tendency of correlative models to ignore sources of biological uncertainty.

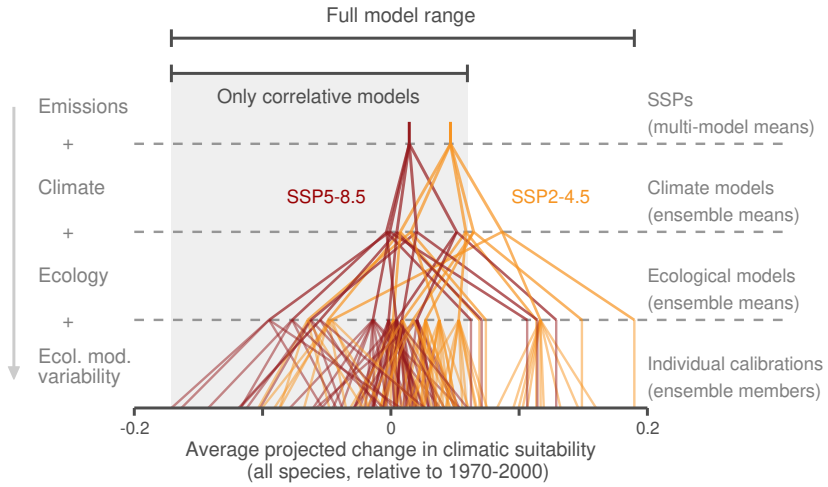


Figure 2: Considering a broad range of models provides a more comprehensive view of possible futures. This figure illustrates the average change in climate suitability across all species and all biomes, with the hierarchical contribution of each source of uncertainty. The top of each cascade represents the overall average change in climate suitability for each Shared Socioeconomic Pathway (SSP). The level below shows the ensemble mean for each climate model (five branches per SSP, corresponding to the five GCMs considered here), averaged across multiple ecological models. The next level represents the contribution of each ecological modeling approach (mechanistic, hybrid, and correlative, i.e., three branches per GCM). The bottom level displays the variation within each approach (e.g., parameter calibrations or statistical algorithms, see Methods). This figure was inspired by the Figure 1.15 in IPCC, 2021: Chapter 1.

An analysis based only on correlative models would have led to an overestimation of the contribution of climate projections (forcing scenarios, climate models, and their two-way interaction) to the total projection uncertainty in all regions, except the Mediterranean. In particular, divergence between climate models would have appeared to contribute as much as ecological models

to projection uncertainty (on average, 36.6% and 37.5%, respectively, Supplementary Figure 2).

More generally, models calibrated using current species distributions—both correlative and hybrid—consistently predicted greater extinction risks than mechanistic models calibrated using empirical data (see Figure 3 for beech; results for the other species are available in Supplementary Figures 3 and 4). For example, correlative models predicted an average loss of 30.5% of current beech distribution (Supplementary Table 1), while mechanistic models led to a much smaller loss (4.48%).

Correlative and hybrid models may be more sensitive to climate change because current distribution data for forest tree species capture only a portion of the climatic niche of a species, underestimating the range of conditions under which it could survive[27,28]. In addition, the current distribution of forest tree species in Europe has been heavily impacted by centuries of active management practices, and these species are unlikely to be in equilibrium with their bioclimatic conditions.

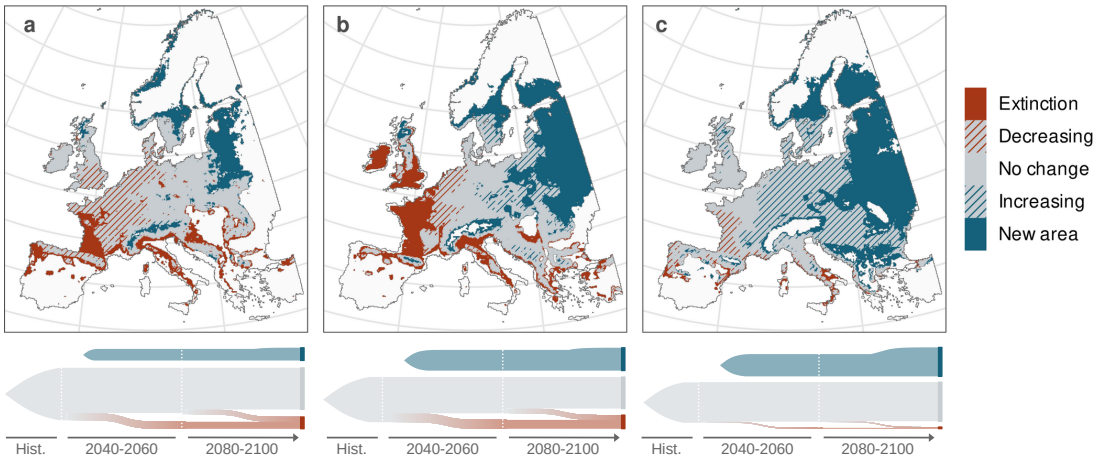


Figure 3: Correlative models project higher extinction risks. Average change in projected distribution for beech (*Fagus sylvatica*) under scenario SSP2-4.5 (2080-2100 relative to 1970-2000), for three ecological modeling approaches: **(a)** correlative, **(b)** hybrid and **(c)** mechanistic models. The maps show the projected change in distribution. The Sankey diagrams below each map illustrate the temporal shifts of distribution. At each time interval, the width of the flows is proportional to the spatial extent where the species has gone extinct, persists, or expands.

Forest management decision-making in an uncertain world

Differences between model predictions can significantly alter country-level projections and impact national strategies. Despite known differences between correlative and other ecological model projections[13–15,17,19], correlative models are routinely used on their own to inform forest adaptation planning, for example through tools such as ClimEssences in France and the Forest Ecosystem Atlas in the United States. Relying on a narrow set of models provides an incomplete picture and can bias decision making[22]. For example, high rates of local extinction predicted by correlative models could guide the decision of intensive intervention strategies, such as large clear cuts followed by introduction of supposedly better adapted exotic species.

Some areas had consistent projections across models and scenarios, and in these areas, management strategies could be implemented with high confidence (Figure 4). For example, around the Mediterranean Basin, models consistently predicted less favorable climatic conditions, for all species considered. In this area, managers may consider a range of adaptation strategies[29], including the introduction of heat- and drought-tolerant species, but making mechanistically grounded predictions for these species will require more data and might be challenging. Similarly, in the boreal biome, the climatic suitability of most species is projected to increase re-

gardless of the model used (Figure 4). This biome includes Finland and Sweden, two important forest countries in terms of wood stock, added value, and forest-based workforce, with forests dominated by two conifer species of important commercial value, Scots pine and Norway spruce. The uncertainty in the future climate suitability of Scots pine was high and the lack of empirical data prevented us from making mechanistic model projections for Norway spruce. Nevertheless, models consistently project that temperate deciduous trees will become more competitive at the northern margin of their range (Figure 3), and prolonged growing seasons in future climates could offer the opportunity to convert pure coniferous stands into mixed forests, to increase their resilience[30,31]. Finally, along the Atlantic margin, the climate suitability of most species is also projected to decrease, except for the two Mediterranean species (pubescent and evergreen oaks), which may replace less adapted species, such as beech[32].

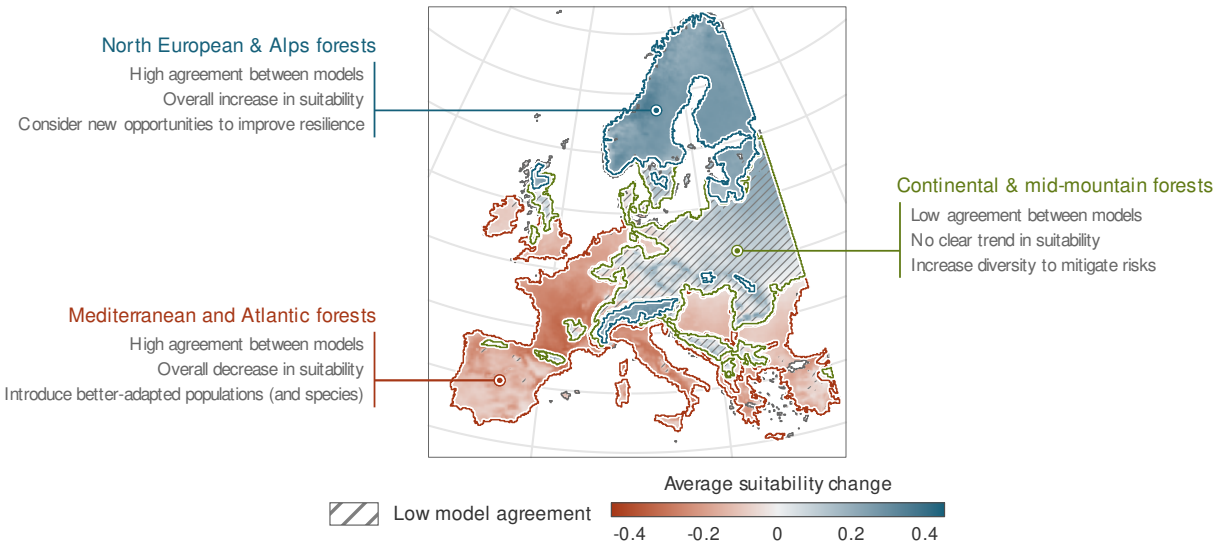


Figure 4: Mapping uncertainties in climate suitability for forest trees across Europe. Three main areas are identified that differ in terms of uncertainty levels, future climate risks and levers of action to address them. The map shows the average climate suitability change across climate models, ecological models and species. Dashed areas highlight regions where ecological models do not agree on the sign of suitability change.

Other areas had inconsistent projections across models and scenarios. In these areas, management has to deal with high level of uncertainty calling for cautious and diverse strategies. Those areas are mostly situated in the continental biome (Figure 4), and in mountainous regions at the transition between Mediterranean and continental/Atlantic climates (Pyrenees, Massif Central, Balkans). In all these regions, a key lever of management action would be the diversification of tree species, as well as the increase in the genetic diversity of populations, to mitigate the risks associated with uncertain future conditions[33–36]. In parallel, more diverse forest management practices, such as promoting uneven-aged stands, could also enhance forest stability by improving structural complexity and buffering against climate extremes[37,38].

Despite the overall highly uncertain projections in the continental biome, projections for some particular species are notably consistent. For example, all models agree on a lower suitability for Scots pines, a commercially important species for Central Europe and the dominant species in more than 30% of forests (Supplementary Figure 1). Projections also suggest that temperate deciduous species—beech and deciduous oaks—will be less affected by climate change, in spite of high uncertainties. Less pessimistic predictions from mechanistic models (Figure 3) suggest that these species could cope with climate change if managers adopt certain adaptive management actions, such as a decrease in stand density to limit competition for water[39]. In addition, maintaining the existing standing genetic variation could help maintain local populations and could be incorporated into genetic enrichment plans[40].

The greater consistency of projections in the Mediterranean and boreal biomes likely reflects the fact that these regions correspond to opposite margins of species' climatic niches, where temperature or drought act as strong limiting factors. In the Mediterranean biome, future warming consistently lowers species performance across models, while in the boreal biome it consistently increases species performance (Supplementary Figure 5). In the continental biome, the historical climate is closer to optimal growing conditions compared to the Mediterranean and boreal biomes (Supplementary Figure 5). Consequently, warming may have diverging effect on the suitability of climate whether populations are moving away or towards their thermal optimum. This generates a higher level of inconsistency between model forecasts, even when estimations of species thermal optima are only slightly different.

The greater consistency of projections in the Mediterranean and boreal biomes likely reflects the fact that these regions correspond to opposite margins of species' climatic niches, where drought and low temperatures, respectively, act as strong limiting factors. In the Mediterranean biome, future warming consistently lowers species suitability across models, whereas in the boreal biome it consistently increases species suitability (Supplementary Figure 5). In contrast, the historical climate in the continental biome is generally closer to optimal growing conditions compared to the Mediterranean and boreal biomes (Supplementary Figure 5). Consequently, warming may either increase or decrease climatic suitability depending on whether populations move towards or away from their optimum. Even small differences in the estimated optimum among models can therefore generate a higher level of inconsistency between model forecasts.

Together, these results contribute to a more complete view of both the threats and opportunities facing forests in Europe, and highlight regions where policies have to be carefully tailored to effectively address all uncertainties. Managers will need to consider the full range of adaptation levers in the face of these uncertainties, including stand management, natural regeneration, and diversification strategies, while considering the risks associated with introducing non-native species[41].

Advancing on two fronts: moving forward with uncertainty while improving ecological models

Large uncertainties in forest tree species distributions by the end of the century demonstrate that improvements in ecological models are needed to provide useful projections to managers and policy makers. Correlative models are becoming more mechanistic through improved mathematical representations and the integration of empirical data[42].In parallel, next-generation mechanistic models will likely integrate more components, including precise phenology, physiological responses, and mortality mechanisms related to drought, pests and fire[43,44]. As these models become more complex, deep learning will offer a promising approach to enhance their predictions while preserving process-based equations[45], with a lower computational cost that could facilitate a more comprehensive estimation of uncertainties. In the near future, these breakthroughs will improve our ability to obtain more robust ecological forecasts, and could in turn reveal new adaptive pathways for forest management. However, these improvements should not prevent us from providing a better framework to identify uncertainty in projections that can guide management today, such as the framework we present here.

The implications of our results extend beyond European forests. Mechanistic models have also been developed for North America and Asia[46,47]. Their projections could be systematically integrated into a comprehensive framework such as the one we present here, alongside correlative model projections. Continental-scale uncertainty assessments, going beyond regional analyses[48], would provide more robust guidance for forest management This is particularly critical in countries where forests are managed at a broader scale than in Europe, such as the United States, and where it could thus be easier to incorporate uncertainty into large-scale decision making.

Given the rapid pace of climate change, debating over which modeling approach to favor

may be less effective than bridging diverse methodologies to generate practical insights and support evidence-based decision making. Ensuring transparency about projection uncertainties is key if we expect forest managers to acknowledge and integrate them[24]. This is how ecological modeling can currently become truly relevant to decision making.

Methods

Species distribution models

This study encompasses a diversity of species distribution models, with three different approaches: correlative models, mechanistic models and hybrid models—i.e. mechanistic models calibrated like correlative models[49].These models were previously evaluated by comparing their projections over thousands of years[19].

We selected four well-established correlative models[50]: generalized linear model (GLM) with lasso regularization, generalized additive models (GAM), boosted regression trees (BRT), and down-sampled Random Forest. The tuning of each model followed a previous study in which their predictive performance was extensively evaluated[50]. We also applied the mechanistic model PHENOFIT which has been validated for several North American and European species, both in the historical and in the Holocene climatic conditions[46,51–53,19]. For the hybrid approach, we calibrated PHENOFIT using the same species occurrence data as correlative models[49]. We optimized the parameters of the model using the covariance matrix adaptation evolution strategy[54] (CMA-ES).

Climate projections

Future simulations were run with the last Coupled Model Intercomparison Project Phase 6 (CMIP6) climate projections, for 5 global climate models (GCMs) and 2 shared socio-economic pathways (SSPs). We used model projections that were downscaled to a 0.1° resolution with a statistical trend-preserving method (the cumulative distribution function transform), using the ERA5-Land reanalysis as a reference observational dataset between 1981 and 2010[55]. We used the projections of the five GCMs considered as good representatives of the full CMIP6 ensemble[55] (GFDL-ESM4[56], IPSL-CM6A-LR[57], MPI-ESM1-2-HR[58], MRI-ESM2-0[59] and UKESM1-0-LL[60]).

Uncertainty partitioning

Our approach was inspired by the partitioning of uncertainties in climate projections originally developed by Hawkins and Sutton[61,62], which was subsequently enhanced with additional methodologies[63,64]. Instead of using a simple variance decomposition approach, we performed an ANOVA-based variance decomposition to also estimate the importance of the two-way interaction effects. All statistical analyses were performed in R[65].

Across all species, we partitioned three sources of uncertainty: the climate projection uncertainty related to the different GCMs, SSPs, and their interaction, the species distribution modeling uncertainty related to the different SDMs. We also considered the interactions between SDMs and climate projections (GCMs and SSPs). For each year, the climate suitability of a cell was considered as a 21-year moving average suitability (e.g. 2040-2060 for the year 2050). We then computed the difference of suitability with the historical suitability (c1970-2000). For each GCM and each SSP, when multiple SDM projections were simulated within the same SDM approach (e.g. multiple algorithms for correlative approach), we kept one ensemble per approach. For each year t, we then applied a linear ANOVA to calculate the sums of squares attributable to each uncertainty source:

$$SS_{tot} = SS_{GCM} + SS_{SSP} + SS_{GCM:SSP} + SS_{SDM} + SS_{SDM:GCM} + SS_{SDM:SSP} + SS_{residuals}$$

We then computed 90% uncertainty ranges additively and symmetrically around the mean projection (assuming a gaussian-like error structure), across all GCMs, SSPs, SDMs, e.g. for SDM uncertainty: $\pm 1.645 * \sigma * \frac{SS_{SDM}}{SS_{tot}}$.

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